

MODEL INTERPRETATION REPORT

Project : Advanced Regression Model Comparison & Optimization Platform

1. Business Objective

The objective of this project is to predict residential house prices using machine learning regression algorithms. Accurate predictions assist buyers, sellers, real estate agencies and financial institutions in making better pricing decisions.

2. Dataset Overview

Property	Value
Dataset	House Prices - Advanced Regression
Records	1460
Features	80
Target Variable	SalePrice
Missing Values	Handled
Outliers	Treated
Encoding	Applied
Scaling	Completed

3. Feature Importance

Feature importance measures how much each feature contributes towards predicting the final house price. Higher importance indicates greater influence on the model prediction.

Feature	Importance
Overall Quality	0.513
Living Area	0.213
Basement Area	0.138
Year Built	0.081
Garage Cars	0.046
Full Bathrooms	0.008

4. Feature Interpretation

Overall Quality

Overall quality has the greatest impact on selling price. Better construction quality generally leads to significantly higher property values.

Living Area

Larger living areas provide more usable space and therefore increase the market value of the property.

Basement Area

Houses with larger finished basements generally attract higher selling prices.

Year Built

Newer houses often require less maintenance and include modern amenities, making them more valuable.

Garage Cars

A larger garage improves convenience and contributes positively to property value.

Full Bathrooms

Additional bathrooms improve functionality and slightly increase selling price.

5. Business Insights

- Houses with higher construction quality consistently achieve higher selling prices.
- Increasing living area significantly improves property value.
- Recently constructed houses generally outperform older properties in market price.
- Basement space adds additional market value.
- Garage capacity positively influences buyer decisions.
- Proper preprocessing improved prediction accuracy across all regression models.

6. Prediction Interpretation

The deployed Random Forest Regressor evaluates multiple property characteristics simultaneously before estimating the final selling price. Instead of relying on a single factor, the model combines information from all important housing features to generate a reliable prediction.

7. Business Recommendations

- Improve overall construction quality before selling a property.
- Renovating living spaces can substantially increase resale value.
- Additional garage capacity improves buyer interest.
- Modernizing older homes increases market competitiveness.
- Use the prediction system as a decision-support tool rather than replacing professional property valuation.

8. Advantages of the Selected Model

- High prediction accuracy.
- Handles nonlinear relationships effectively.
- Robust against overfitting.
- Suitable for real-world property valuation applications.
- Performs consistently across different housing characteristics.

9. Conclusion

Based on the experimental evaluation and feature analysis, the Random Forest Regressor was selected as the final model for deployment. The model demonstrated reliable prediction performance and identified Overall Quality and Living Area as the most influential factors affecting residential house prices. The developed system can assist buyers, sellers, real estate professionals and financial institutions in making informed pricing decisions.